

# ARJUN SHARMA

## Software Engineer

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## PROFESSIONAL SUMMARY

Computer Science graduate with a strong foundation in algorithms, system design, and distributed systems. Solved 500+ LeetCode problems (Top 5%, Knight badge). Experienced in building high-performance, fault-tolerant backend systems and scalable full-stack applications. Interned at a product-based company working on real-world, high-traffic systems. Passionate about open-source, low-level optimizations, and engineering at scale.

## WORK EXPERIENCE

### Software Engineering Intern

Google (via STEP Internship Program), Bangalore

May 2024 – Aug 2024

- Owned and shipped a latency optimization for an internal data pipeline (Google-scale), reducing P99 latency by 42% using batching and async processing in Go.
- Designed and implemented a new caching layer using Redis that cut redundant DB reads by 60% across 3 downstream services.
- Wrote comprehensive unit and integration tests (95%+ coverage) and passed code reviews with L5/L6 engineers.
- Presented findings to a cross-functional team of 15 engineers; solution was adopted into production.

### Backend Engineering Intern

Razorpay, Bangalore

Nov 2023 – Jan 2024

- Built a real-time webhook retry mechanism using AWS SQS + Lambda, handling 1M+ events/day with 99.98% delivery reliability.
- Refactored legacy monolith endpoints into microservices (Spring Boot), cutting response time by 28%.
- Identified and fixed a race condition in concurrent payment state updates using optimistic locking; prevented ~\$12K/month in duplicate charge errors.
- Wrote runbooks and internal documentation adopted by 4 team members.

## EDUCATION

### B.Tech in Computer Science and Engineering

Indian Institute of Technology (IIT) Bombay

Graduated: 2021 – 2025

CGPA: 9.2 / 10 | Dean's List (All Semesters) | Relevant Coursework: Advanced Algorithms, Operating Systems, Distributed Systems, Database Internals, Computer Architecture, Compiler Design, Computer Networks

## SKILLS

**Technical Skills:** Python, C++, Java, JavaScript (ES6+), TypeScript, Go (Basics), SQL, Data Structures & Algorithms (Advanced), System Design (LLD + HLD), React.js, Node.js, Spring Boot, REST APIs, gRPC, GraphQL

**Databases & Storage:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB (Basics), Cassandra (Concepts)

**DevOps & Cloud:** AWS (EC2, S3, Lambda, RDS, SQS, CloudWatch), Docker, Kubernetes (Basics), Terraform (Basics), CI/CD (GitHub Actions, Jenkins), Linux, Bash Scripting

**CS Fundamentals:** Operating Systems, Computer Networks, DBMS, OOP, Distributed Systems, Concurrency & Multithreading, Memory Management

**AI & Tools:** OpenAI API, LangChain, Git, Postman, Jira, Figma, IntelliJ, VS Code

**Soft Skills:** First-Principles Thinking, Ownership Mindset, Clear Technical Communication, Collaborative Problem Solving, Handling Ambiguity, Attention to Scale

## PROJECTS

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- Distributed Key-Value Store | C++, Raft Consensus, gRPC, Docker | Mar 2025
- Built a fault-tolerant, distributed KV store from scratch implementing the Raft consensus algorithm with leader election, log replication, and snapshotting.
- Achieved linearizable reads/writes with <10ms latency under 10K concurrent clients in benchmarks.
- Inspired by MIT 6.824 (Distributed Systems); passed all 50 provided test cases including network partitions.
- GitHub: [github.com/arjunsharma/raft-kv-store](https://github.com/arjunsharma/raft-kv-store)
- Mini Search Engine | Python, Inverted Index, TF-IDF, FastAPI, React.js | Dec 2024
- Designed and built a full-text search engine from scratch with web crawling, tokenization, inverted indexing, and TF-IDF ranking over 100K+ documents.
- Achieved sub-50ms query response time; deployed on AWS EC2 with load balancing.
- GitHub: [github.com/arjunsharma/mini-search-engine](https://github.com/arjunsharma/mini-search-engine)
- URL Shortener at Scale | Go, Redis, PostgreSQL, Docker, Kubernetes | Sep 2024
- Designed a system capable of handling 100K redirects/second using consistent hashing, Redis caching, and horizontal scaling via Kubernetes.
- Wrote a detailed HLD/LLD document covering capacity estimation, DB sharding, and CDN integration (received 400+ stars on GitHub).
- GitHub: [github.com/arjunsharma/url-shortener-scale](https://github.com/arjunsharma/url-shortener-scale)

## ADDITIONAL INFORMATION

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**Languages:** English (Fluent), Hindi (Native), Marathi (Native)

**Competitive Programming:** LeetCode – 500+ solved, Knight Badge, Top 5% globally | Codeforces – Specialist (1450+) | Google Kickstart 2024 – Global Rank 312 | ICPC Regionals 2023 – Qualified & Ranked Top 50 (Asia-West)

**Certifications:** AWS Solutions Architect – Associate (2024), Meta Backend Developer Certificate (Coursera, 2024)

**Open Source:** Contributor to facebook/react (2 merged PRs – performance fix & docs), redis/redis (issue triage & fix)

**Achievements:** Smart India Hackathon 2024 – National Winner | Institute Technical Council – Core Team Lead | Published research paper on "Optimizing Cold Start Latency in Serverless Architectures" – IEEE Xplore 2025

**Interests:** Systems Programming, Open Source, Competitive Programming, Tech Writing (Medium – 5K followers)